me to strike a barter deal with the French, saying Airbus was selected expecting the French to include in the Viking contract the stage engineering technology?' Dhawan said no.

'These are two different matters, Mr Nambi,' he said. 'And don't waste your time waiting for my decision on the aircraft; you go ahead with your job of making the rocket engine.'

Finally, Dhawan selected Airbus, and I am sure he remained untouched or unmindful of the fact that he just let go of an opportunity to get associated Viking technology without much toil. It is another matter that we learned and developed all these associated stage technology, though at the cost of more time and energy.

Almost ten years later, here we were after the successful ground test of Vikas, with the mandate to put together the 'flying' engine, and second stage for PSLV. During our stay in Vernon, some of the team members had learned quite a bit on these subsystems though it fell outside the purview of the contract, but now we were called to demonstrate and deliver.

We sat down to identify the tasks. The first crucial decision was to not use any indigenous parts in the first flight. In other words, we would fabricate bigger parts with the same materials and processes that went into the making of Viking by the French, or use smaller parts like fasteners and bearings fabricated by the original SEP suppliers. This was to minimise risks and focus on our technological mastery over putting together the engine than on fabrication. Our other acquired skills, including those material development and fabrication could be put to test after three successful flights of PSLV, it was decided.

I got my colleague G Krishnamurthy, a mechanical engineer specialized in fabrication, to compile a materials manual. This involved making a list of materials used in Viking/Vikas, with details of their Indian and American equivalents. This would be an exhaustive list that would include the chemical composition and physical properties of the materials, including their tensile strength, durability and fatigue, besides expected behaviour under varying conditions.